

RTG 2947 AirMetro



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Wireless Communication Networks in Advanced Air Mobility (AAM) : Pre-flight

PhD Talk, Friday , 28.06.2024

Use- Cases in AAM – „PURPOSE”

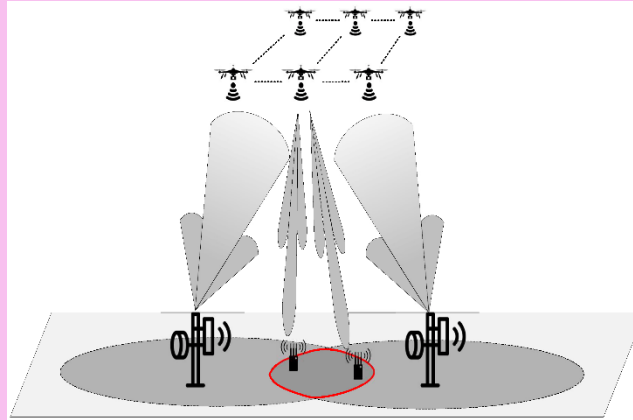
Surveillance



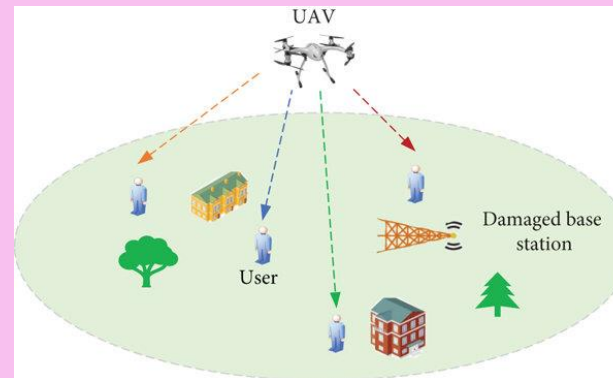
Drone Delivery



Network expansion



Network establishment



Passenger transport



Vertiport

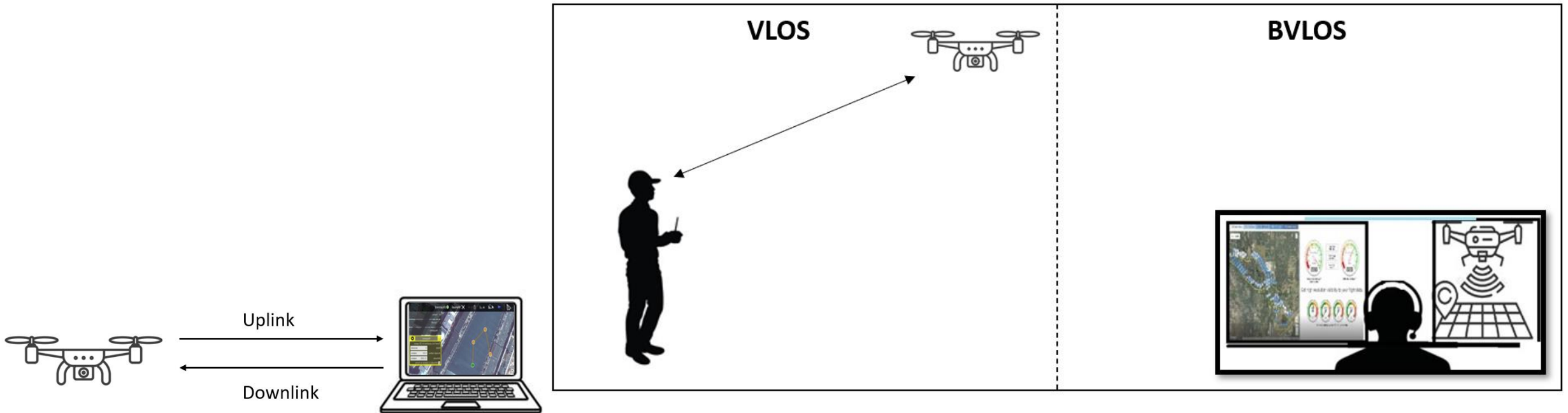


Wireless communication is a key for safe Beyond Visual Line-of-sight operations

[1] <https://images.google.com/> , drone cartoon preview images <https://alamy.com>

Flight Information - „HISTORY“

- M.Sc. Communications Engineering , TU Munich
- 2 years research student at Atmosphere GmbH , weßling (Communication for Unmanned Vehicles)



Uplink – Telemetry message (current altitude, GPS, battery,..)

Downlink – Navigation commands (waypoints, steering)

VLOS Visual Line of Sight BVLOS Beyond Visual Line of Sight

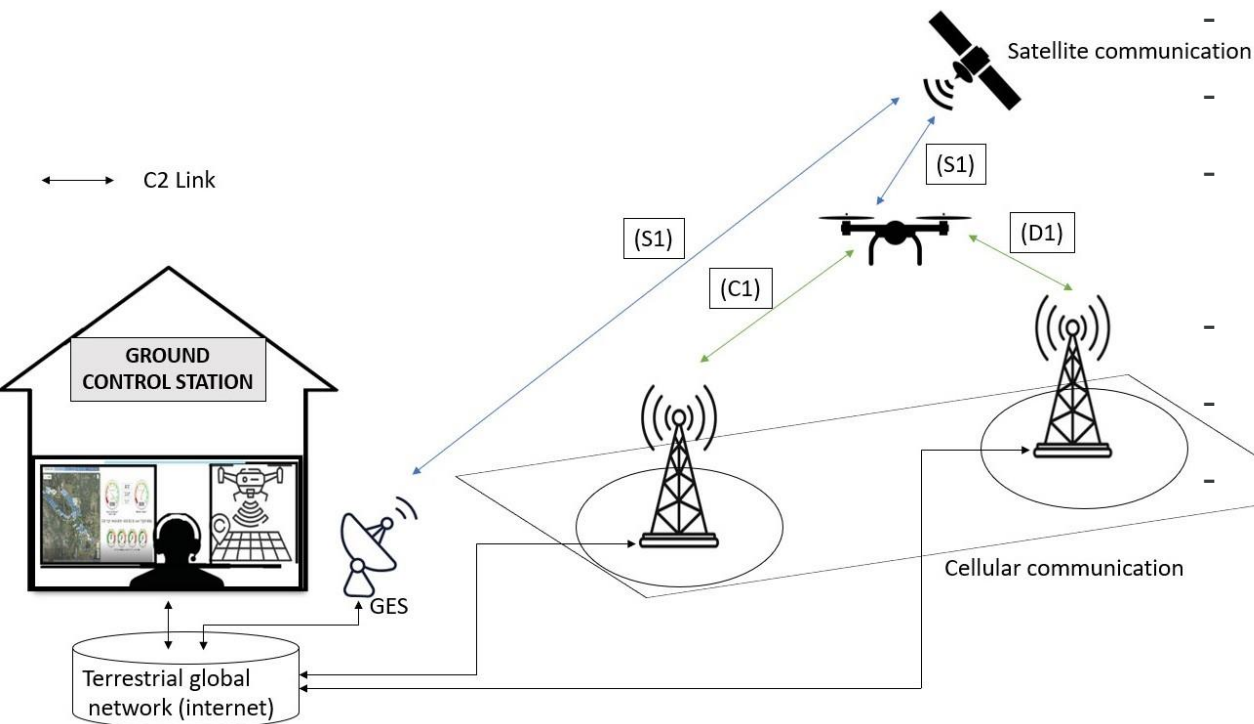
[2] ICAO International Civil Aviation Organization, ICAO, “Performance-based Communication and Surveillance (PBCS) Manual”, 2017

MAVLink Protocol Enhancement for Command and Control (C2) communication in integrated multi-link scenario

Supervisors : Jörg von Mankowski, Thomas Wöllert, Dr. Markus Werner, Prof. Wolfgang Kellerer

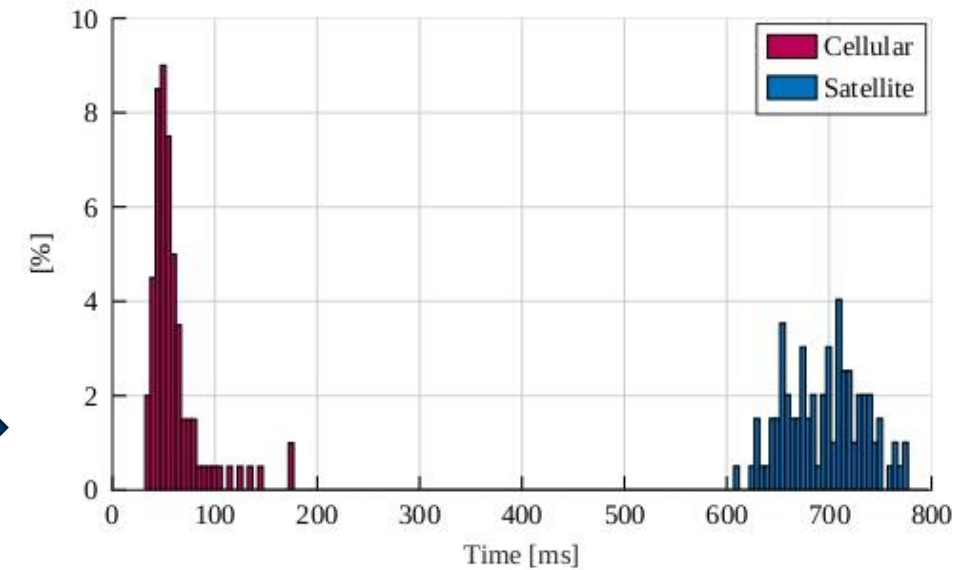
External : Atmosphere GmbH , Lehrstuhl für Kommunikationsnetze (TUM)

"Availability alone does not guarantee safe and reliable communication"



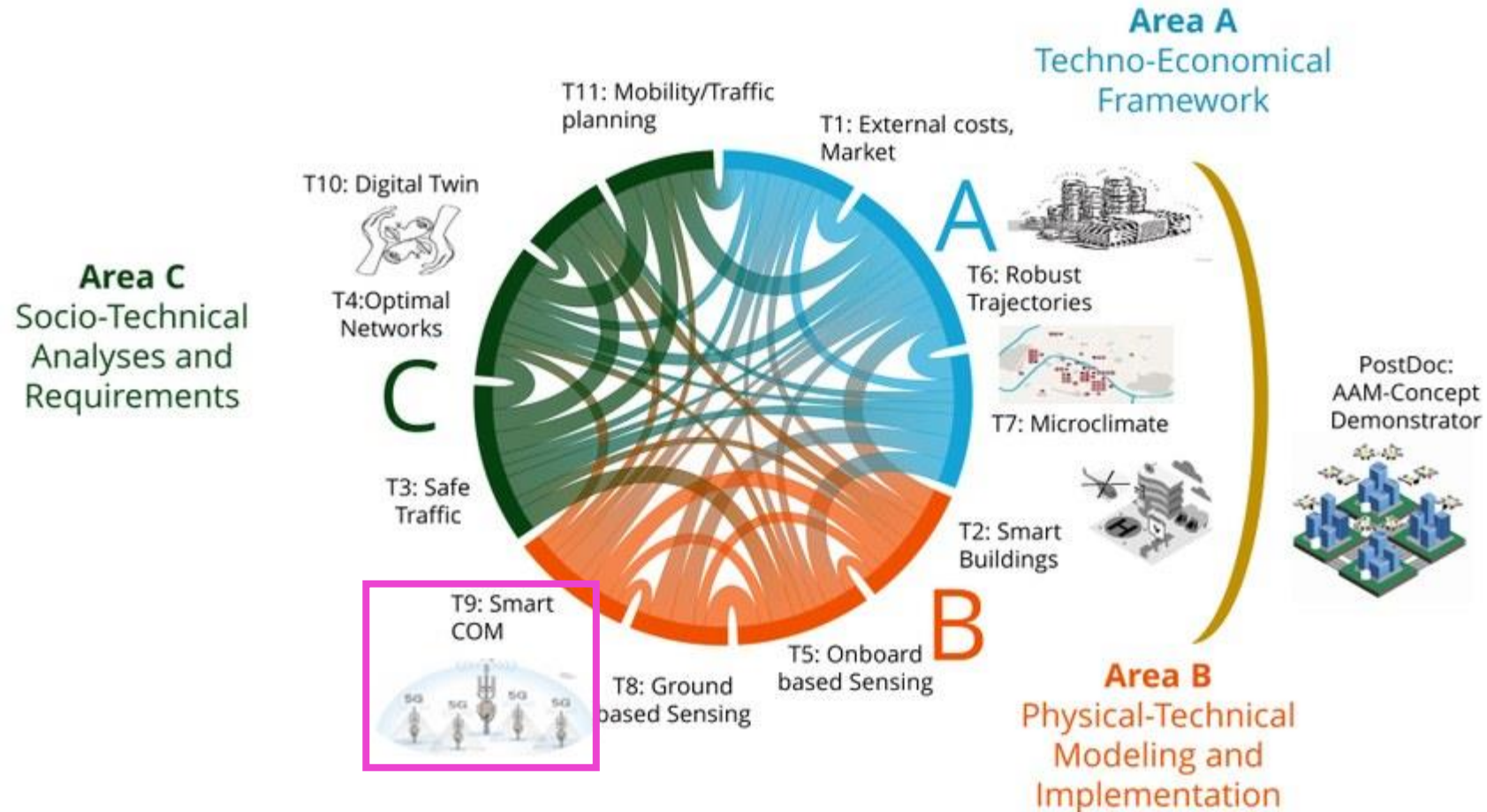
- Integrated multi-link for reliable C2 communication
- 5G – Primary , Iridium – Secondary (backbone)
- ICAO C2 Performance recommendation [2]
- Reliable = {Availability, Continuity, Transaction time, Integrity}
- **Logical Link Characteristics (Link Bandwidth, Latency)**
- Evaluate the attributes on MAVLink telemetry protocol
- Link quality matrix for reliable and safe operations

Attribute	5G	Iridium
Data transmission	Consistent	Consistent
Bandwidth Allocation	Fast	Noticeable delay
Bandwidth	Satisfy above 100 kbps	22kbps / 88 Kbps
Burst arrival	None	Present
Latency	Negligible (~ 50 msec)	Significant (750 msec)

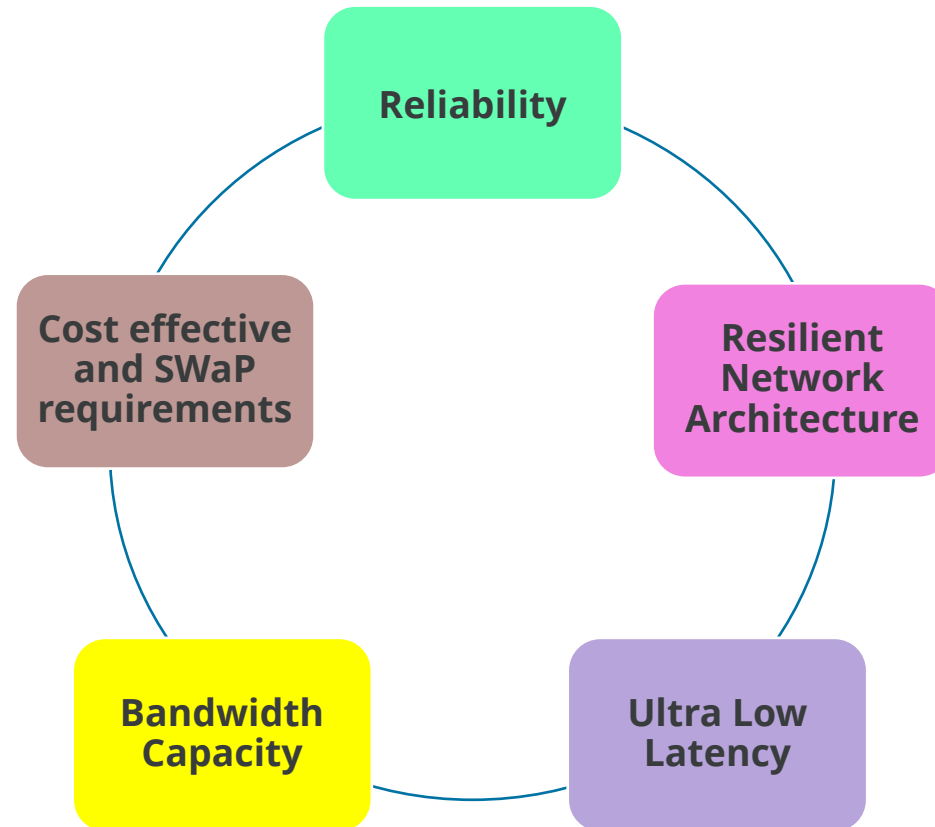




- Smart Mobility Lab at Hoyerswerda



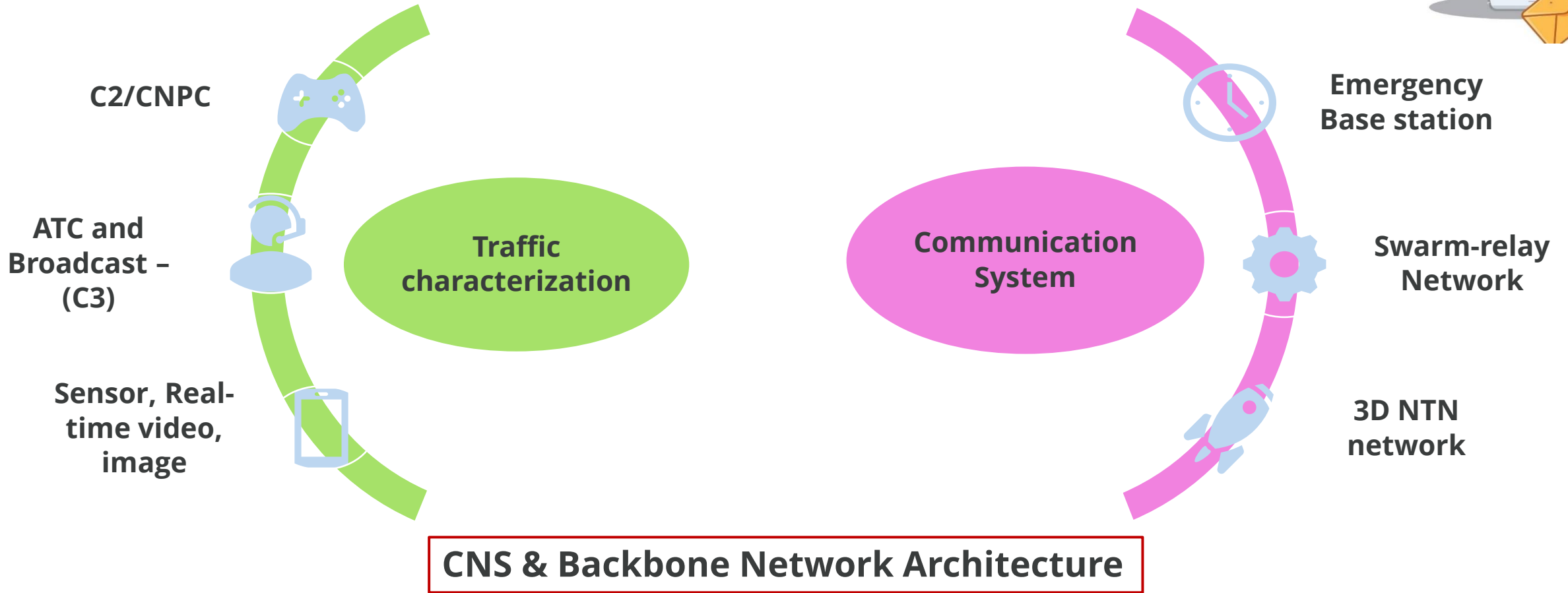
Wireless communication – „BRIEFING“



Terrestrial Communication – Cellular (LTE, B5G)

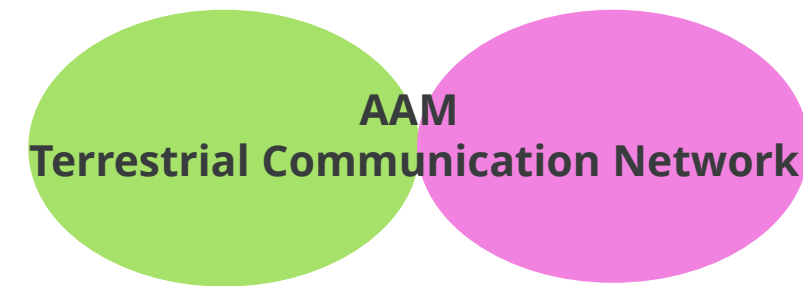
SWaP Size, Weight, and Power

Terrestrial Communication Network in AAM – „REQUISITE BULLETIN“



C2	Command and Control
CNPC	Command and Non-Payload Communication
C3	Command, Control and Communication
NTN	Non-terrestrial Networks
CNS	Communication, Navigation and Surveillance

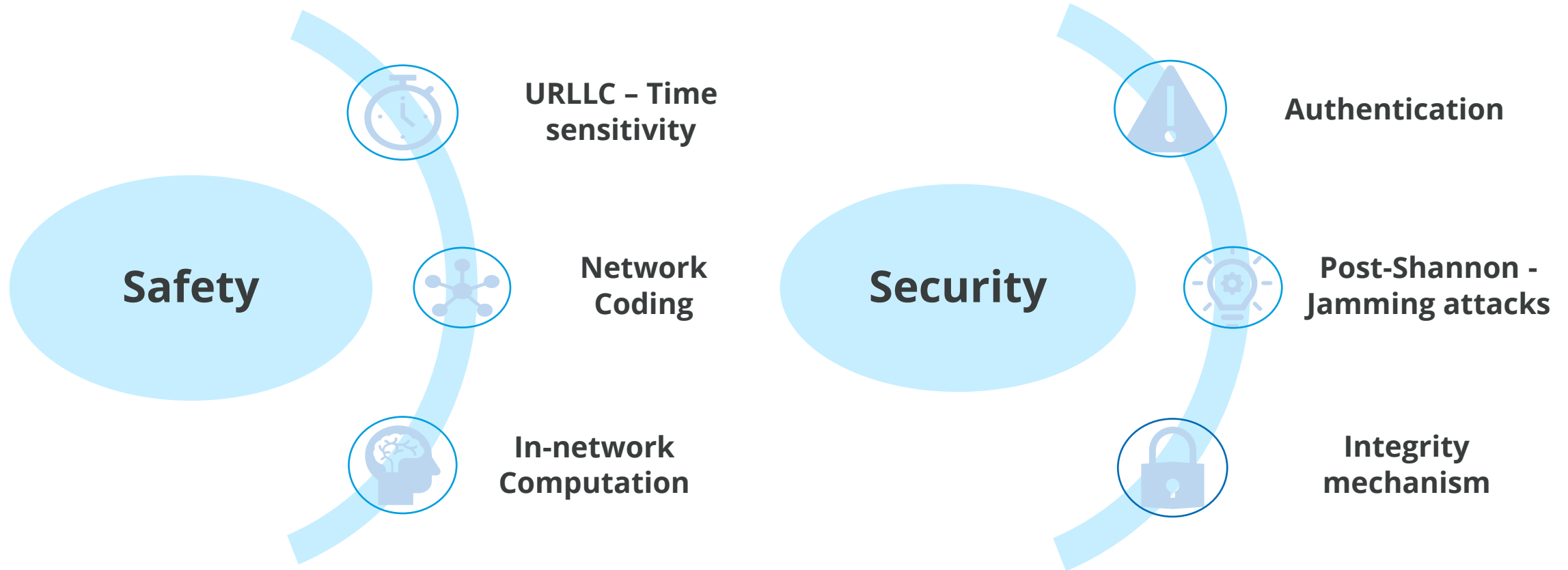
Build AAM communication – „ FLIGHT PLAN“



Communication and Computation for flight operations

AAM Communication Network – „PATH“

WIRELESS MESH NETWORKS



- Reliable and resilient network architecture
- Coding and computation - optimization for ensuring reliability
- Build the Network based on use-case perspective



Thank you for your Attention!

Literature

- SESAR Droc2om project
- 3GPP Rel 17 +
- ICAO SAPRPS
- EUROCAE Technical operational std.
- U-Space and Urban Air Mobility

I/O :

T5+T8 (Nav) , T2 – signal strength +Comm system ,

T10 – digital twin (cyber physical environment)

T6 - Trajectory dependence

T1 – Cost and Market

T11 – Mobility and Traffic planning

T4 – Optimal Path and Network for Landing/take-off sites

